

LEZIONE 28 – 05

MACHINE LEARNING FOR THE ANALYSIS AND CLASSIFICATION OF HUMAN MOVEMENT.

There are many wearable sensors able to measure different parameters:



We talk about **WEARABLE INERTIAL SENSORS**.

- Smartwatch: records human movement
- Smartphone: they embed motion sensors.
- Commercial and prototypes sensors: some of them we buy for other purposes.



Example.

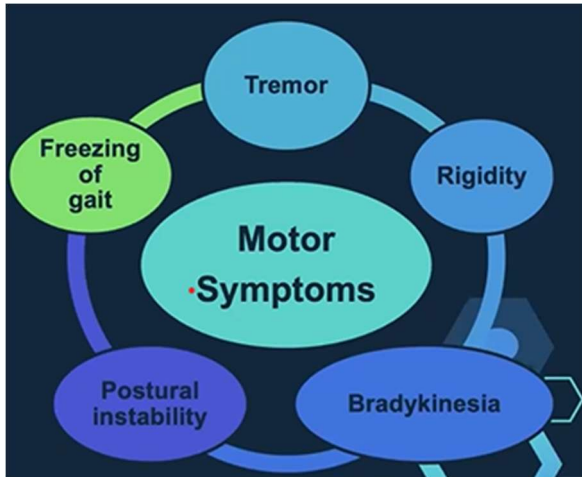
APPLICATIONS:

1. Recognize activities: important when studying movement disorders, like Parkinson's disease.
2. Sport
3. Rehabilitation
4. Well-being.
5. Diagnosis
6. Monitoring of neurological disorders.

MONITORING of PARKINSON'S DISEASE.

250.000 subjects in Italy.

It's the death of dopaminergic neurons, compromises motor control, behaviour, motivation, cognition, sleep, attention and mood.



Tremor is the most common.

Rigidity of muscles.

Bradykinesia: slowness of movement.

Postural instability: increases the probability of falls.

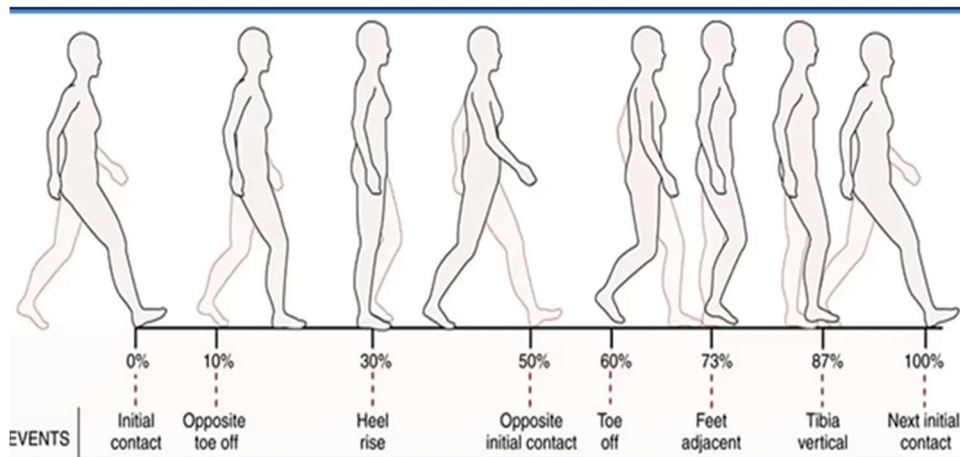
Freezing of gait: sudden motor-block.

Wearable technologies can bring objective evaluations and monitors diseases during daily life.

GAIT ANALYSIS

The walking activity is automated in humans. It requires so much coordination. Gait analysis gets information for **evaluation of the posture stability**.

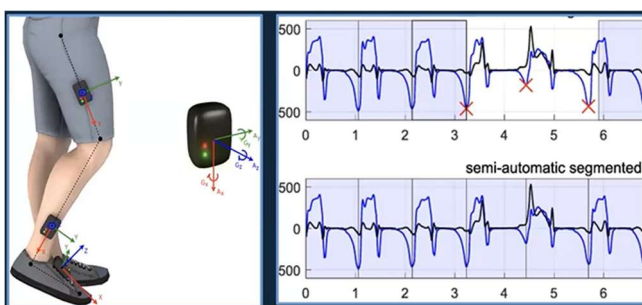
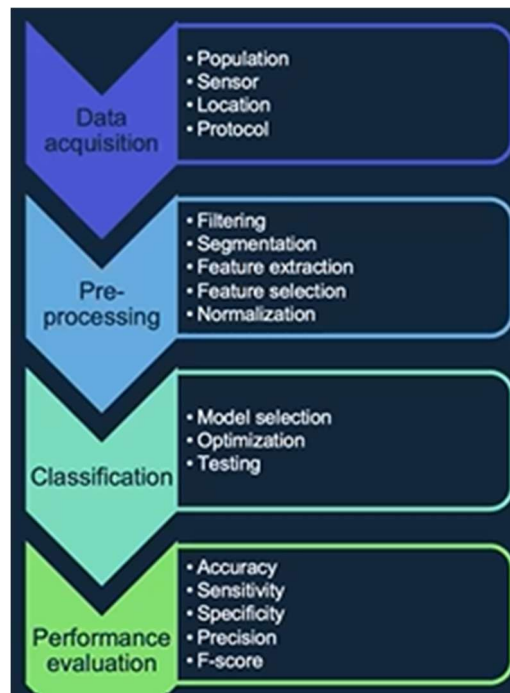
Gait phases begin with the contact of foot on the ground. Initially only the **heel** (*tallone*), after **all** the foot.



We can get information:

- How much time the foot is in contact with the floor.
- How much time the foot is not in contact with the floor.

People with posture problems prefer to have the foot in contact with floor for as much as possible.



The wearable systems allow to study the movement for long period of time.

The **feature extraction** can happen by:

- Expert-knowledge
- Direct observation
- Literature

The **feature selection** can be:

- A-priori. It is **FAST**, but **do not** maximize performance.
- Independent of the model
- Independent of the performance.

It is used **to maximize performance** but is also:

- risk of over-fitting
- time consuming.

DIMENSIONALITY REDUCTION: combining features in other features
(*optimization*). It is:

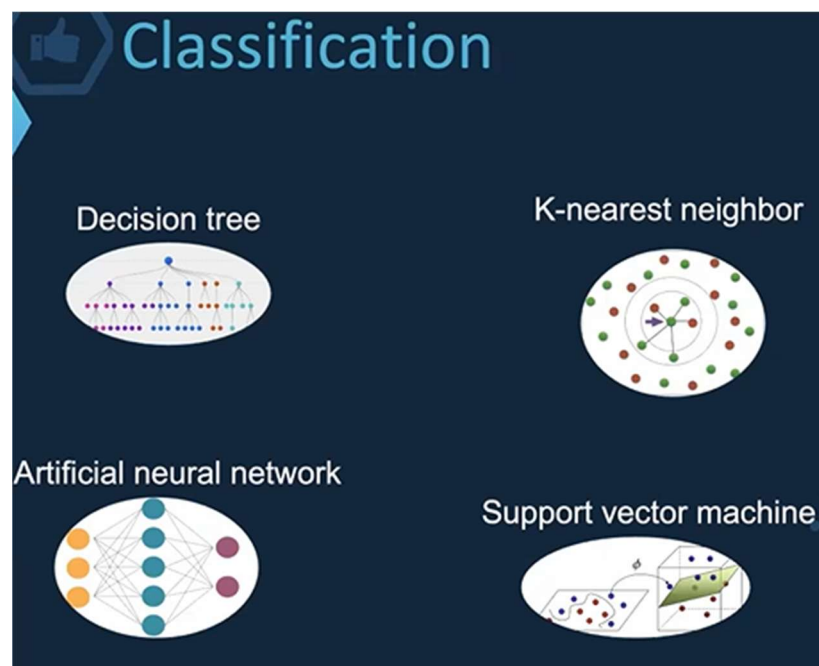
- independent of the target
- reduce the dimension
- small loss of information.

NORMALIZATION:

It's about working with values of different scales:

$f' = \frac{f - \min(f)}{\max(f) - \min(f)}$					$f' = \frac{f - \text{mean}(f)}{\text{std}(x)}$				
f1	f2	f3	...	fN	f1	f2	f3	...	fN
0.13	0	0	...	0	-0.7	-1	-1	...	-1.2
0	0.3	0.17	...	0.4	-0.9	-0.4	-0.6	...	-0.3
0.94	0.6	0.64	...	0.6	0.8	0.4	0.4	...	0.3
1	1	1	...	1	0.9	1.2	1.2	...	1.2

We have different **MODELS**:



DECISION TREE:



Every time that the **outlook** is **overact**, the answer is **YES**. It's very related to the target.

K-NEAREST NEIGHBOUR:

- Choose number of neighbours
- Compute the distance from neighbours
- Assign the element to the majority class.

DEEP LEARNING APPROACH.

Requires much data
Requires much computational power
Less interpretable

- Improve performance
- Improve generalization capability
- Reduce user dependency

PART 2

A guide to find skeletons in videos: VISUAL BODY TRACKING.

Analysis of human motion provides immediate insight on alterations of such systems.

Inertial measurement unit: useful to measure *acceleration*, or *rotation* of human body.



IMUs

Motion capture system



EMG sensors

Electrical activity of muscle.



Camera-based
MOCAP

(**GOLDEN STANDARD**)

Motion Capture Systems (I)

Set of video cameras essentially organized inside a laboratory, the person performs motor activities while recorded by cameras. They need to apply some markers on the person.



For the *next generation* VBT system we would something:

- working among in a park, shopping mall, outside a house.
- Low cost
- Markerless.

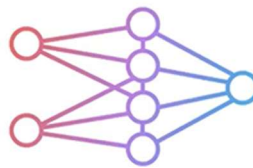
allow to process videos and images with high performances.

Deep-Learning Visual Body Tracking

Starting from RGB video, provide directly frames that compose the video, processed by **deep neural network**. From each frame, it can extract a virtual skeleton reconstruction.



RGB video



Deep Neural Network



Virtual skeleton reconstruction

TOP-DOWN APPROACH:

1. Localize the humans
2. Estimate the body parts
3. Calculate the pose as likely joint connections

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- ✓ Easier to implement
- ✗ Accuracy is upper bounded by human detector
- ✗ Increasing # of subjects to track negatively affects performance

BOTTOM-UP APPROACH:

1. Estimate the human body parts
2. Identify joints-subject mapping
3. Calculate the pose of each subject as likely joint connections

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- ✓ Performance are less affected by # of subjects to track
- ✓ Accuracy is higher on average
- ✗ Complex to implement

Deep Pose

Toshev, Alexander, and Christian Szegedy. "DeepPose: Human pose estimation via deep neural networks." *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2014.

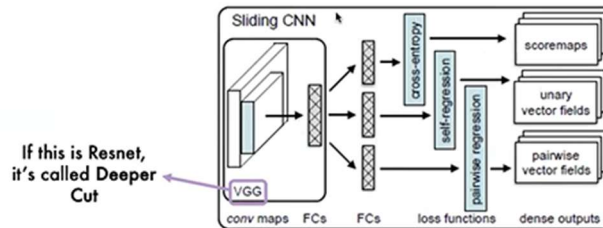


- Joint estimation is formulated as a regression problem, each joint is estimated from convolving the entire input frame
- A cascade of refiner regressors is applied to sub images including the joint location to refine the estimation

It's a bottom-up approach.

Deep Cut

Pishchulin, Leonid, et al. "Deepcut: Joint subset partition and labeling for multi person pose estimation." *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2016.



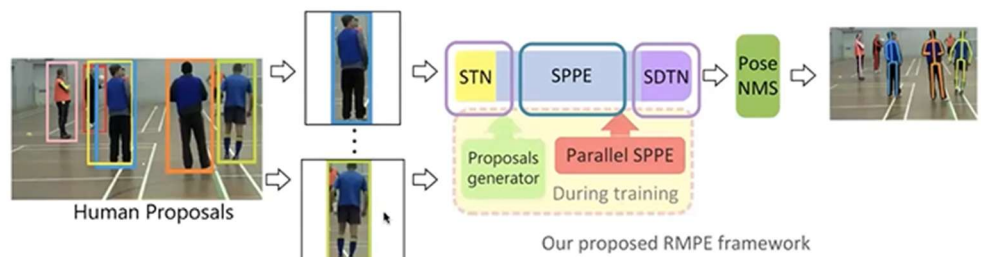
- Detect all possible body parts in the given image, then label them such as a head, hands, legs, etc., then assign them to each subject
- Integer Linear Programming (ILP) modelling to implicitly group all the detected key points in the given input such that the resulting output resembles a skeleton representation of the human.

Deep-Learning VBT



Alpha pose

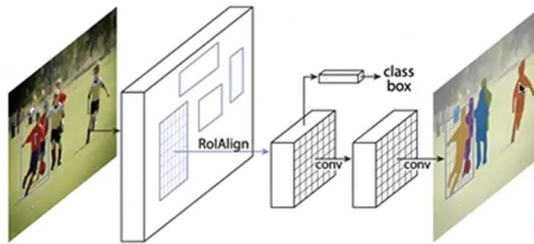
Fang, Hao-Shu, et al. "Rmpe: Regional multi-person pose estimation." *Proceedings of the IEEE international conference on computer vision*. 2017.



- Two problems with top down: incorrect bounding box, multiple detection of same subject
- Two modules are defined to tackle these issues: **Symmetric Spatial Transformer Network (SSTN)** and **Single Person Pose Estimator (SPPE)**

Mask R-CNN

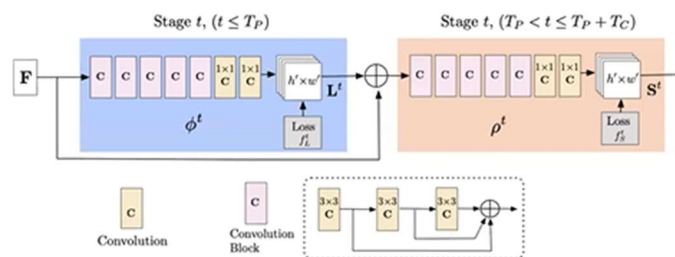
He, Kaiming, et al. "Mask r-cnn." *Proceedings of the IEEE international conference on computer vision*. 2017.



- Employ image segmentation from ROI through a mixture of CNN an RNN to identify humans
- Once the "human" masks have been identified perform pose estimation with any efficient method

Open Pose

Cao, Zhe, et al. "Realtime multi-person 2d pose estimation using part affinity fields." *Proceedings of the IEEE conference on computer vision and pattern recognition*. 2017.



- Open pose was the game changer thanks to accuracy and performance and it is still one of the most powerful solution for 2D tracking
- The first network predicts a set of the confidence map for each body part while the second branch predicts a Part Affinity Fields (PAFs) which creates a degree of association between parts. It is also useful to prune the weaker links in the bipartite graphs.

2D provides limited information and fails in identifying complex poses with joint occlusions.

UPLIFTING PROBLEM: we want to identify a way to move to 2D model to 3D model.

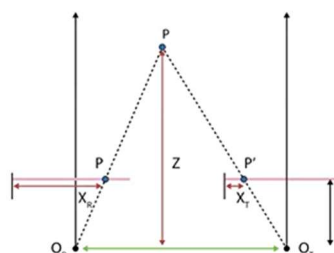
The first way is triangulation of 2D coordinates from multiple cameras.

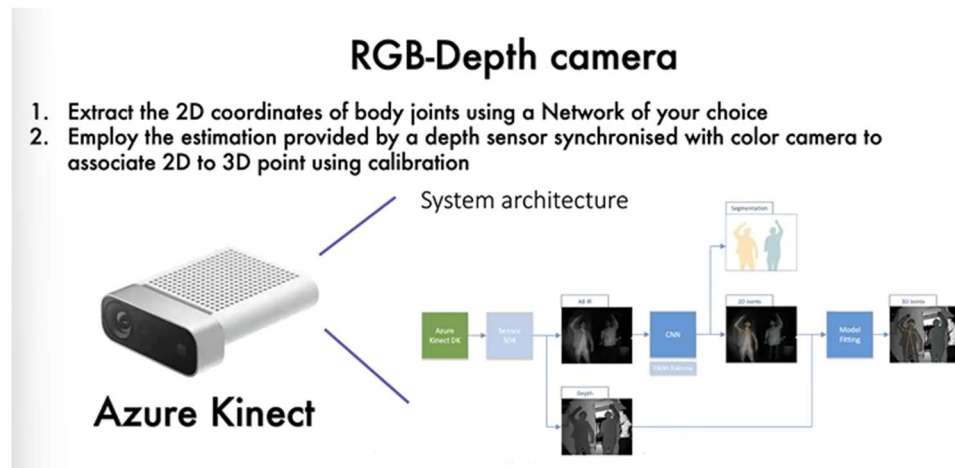
The second is the employ of RGB-Depth cameras (Kinect).

The third is to train the network to estimate directly 3D coordinates.

Triangulation

1. Extract the 2D coordinates of body joints using a Network of your choice from at least two different camera perspectives
2. Triangulate the joint positioning in the 3D space described by the 2 cameras

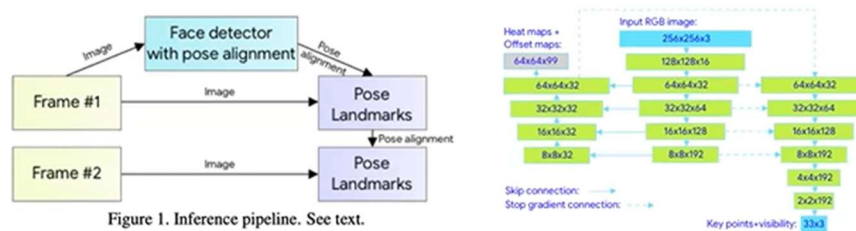




3D Pose Estimation Networks.

Employ a **double stage network**: first stage learns 2D regression, second stage learns how to automatically uplift 2S to 3D from training data projection and reprojection.

Train the whole model directly on monocular videos and images with 3D annotations or synthetic data:
BLAZE POSE, neural network architecture.



- First face is identified, then the remaining joints of the skeleton are identified through heatmaps and regression
- Highly efficient and lightweight (can run on mobile devices in real time at 30 FPS with CPU only!!)
- It is included in Tensorflow pose estimation and Google Mediapipe Library (more at the end...)

Deep-Learning VBT

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The Uplifting problem

- ✓ Many deep learning solutions are possible for 2D and 3D body tracking/pose estimation: choose according to your need (go for Open Pose or Blaze Pose if in doubt)
- ✓ Deep Learning VBT can infer also occluded joint from model estimation, can be applied to in-the-wild videos with good accuracies, with few constraints and without any need for physical markers
- ✗ Accuracy is still limited with respect to MOCAP, in particular for complex poses and occlusions
- ✗ Virtual markers position is fixed by training data, to change markers, retraining is needed
- ✗ In multisubjects tracking, some additional consistency among video frames has to be introduced

VBT for Health applications

Motion is a reservoir of information about a human being:

- alterations of motor functions are typical in the elderly, people with neurological diseases like Parkinson's disease or after severe brain injuries (post stroke).

Gait Analysis

Alterations in how an elderly walks (gait cycle) are strictly related to increase risk of fall and reduced Quality of Life.

Gait Analysis using Azure Kinect

1. The body tracking algorithm of Azure Kinect extracts skeleton of subject walking
2. Hand-crafted features from skeletal model are used to characterize gait objectively
3. Shallow Learning is used to identify alterations in gait patterns



Hand dexterity is a relevant indicator of the level of impairment in many neurological diseases

Hand Tracking



Hand tracking poses greater challenges than body tracking

Smaller tracked surface

Higher density of markers required

Fingers have higher dynamics

GMH-D: Combining Google MediaPipe and Kinect

1. GMH coordinates are uplifted using Kinect depth sensor
2. Hand-crafted features from skeletal model are used to characterize hand dexterity objectively
3. Shallow Learning is used to identify alterations in hand movements

Exergaming



New paradigm of training and rehabilitation in a gamified setting



- Increased user engagement
- Elicitation of specific movements through a goal-oriented setting
- Acquisition of additional data for evaluating patient's health status

Exergame - Lower Limb Rehabilitation

The body tracking algorithm running in real-time allows the control of the virtual avatar